Pediatric Failed Airway

AR-06 | Airway & Respiratory

Quick flow

01 Declare Failed Airway [EMT / AEMT / Paramedic]

Cannot oxygenate/ventilate • unsafe anatomy • deterioration • or three failed laryngoscopy attempts.

02 Call + Optimize BVM [EMT / AEMT / Paramedic]

Bring most experienced provider/rescue equipment; position • suction • adjuncts • two-person seal • oxygen • PEEP.

03 BVM Effective? [EMT / AEMT / Paramedic]

Judge chest rise, SpO_2 trend, mental status/perfusion, and $EtCO_2$ when available.

04 Rescue Airway Needed? [AEMT / Paramedic]

If BVM ineffective or prolonged ventilation required: authorized age/size-appropriate BIAD or deliberate paramedic rescue.

05 Cannot Oxygenate / Ventilate? [Paramedic]

Paramedic considers age-appropriate, specifically authorized surgical rescue only after less-invasive rescue fails or cannot be used.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Confirm + Transport [EMT / AEMT / Paramedic]

Waveform $EtCO_2$ • secure device • maintain warmth • post-airway care • immediate notification.

Overview

- Declare pediatric airway failure early, stop repeated laryngoscopy, and return immediately to effective BVM ventilation.
- Effective BVM is an acceptable definitive field airway. Use a rescue BIAD when appropriate; cricothyrotomy is a rare final rescue based on age and agency authorization.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Stop further laryngoscopy attempts, call for resources, optimize two-person BVM with positioning, adjuncts, suction, oxygen, and PEEP when indicated.
- Continue BVM as the definitive field airway when oxygenation and ventilation are effective; begin transport without delay.

AEMT

- Perform all EMT care plus authorized pediatric BIAD and continuous waveform capnography. Endotracheal intubation and video laryngoscopy are not included as AEMT procedures in this protocol.
- Use an age/size-appropriate BIAD only when BVM is ineffective or prolonged ventilation requires it; do not make repeated blind rescue-device attempts.

Paramedic

- Perform all prior care plus an age/size-appropriate BIAD or a changed, deliberate video-laryngoscopy rescue approach only when BVM is ineffective or prolonged ventilation requires it.
- Consider age-appropriate cricothyrotomy only for persistent cannot-oxygenate/cannot-ventilate failure after less-invasive rescue fails or cannot be used, and only when specifically authorized.

Indications

- Unable to oxygenate or ventilate, anatomy unsafe for further attempts, or three total unsuccessful laryngoscopy attempts.
- Pediatric deterioration during or after an unsuccessful advanced-airway attempt when continued laryngoscopy could worsen hypoxemia or interrupt ventilation.

Treatment steps

- Declare failure early. Do not wait for three attempts when BVM is ineffective, oxygenation/ventilation is deteriorating, or anatomy makes further attempts unsafe.
- Stop laryngoscopy, bring the most experienced available airway provider and rescue equipment, and optimize BVM with neutral positioning, suction, OPA/NPA when appropriate, correct mask size/seal, oxygen, and PEEP when indicated.
- Judge BVM success by chest rise, SpO_2 trend, mental status/perfusion, and waveform $EtCO_2$ when available - not SpO_2 alone.
- If BVM is effective, continue it as the definitive field airway and begin transport. Do not pursue an advanced airway solely because BVM is not ideal.
- Do not use repeated blind BIAD or laryngoscopy attempts. Every additional attempt must have a meaningful change in provider, approach, positioning, or equipment.
- Confirm every BIAD/ETT with continuous waveform $EtCO_2$, chest examination, SpO_2, depth, and securement; reassess after every movement.
- If heart rate remains below 60/min with poor perfusion despite effective oxygenation and ventilation, start CPR and follow PC-02. If pulseless, follow PC-04 Pediatric Pulseless Arrest.
- Maintain warmth, frequent reassessment, rapid transport, and early notification for any rescue airway, persistent hypoxemia, or respiratory failure.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Warnings / precautions

- Effective BVM ventilation is an acceptable definitive field airway; avoid repeated attempts that interrupt oxygenation or worsen hypoxemia.
- No more than three total laryngoscopy attempts; every further attempt must change the provider, approach, positioning, or equipment.
- Do not use repeated blind BIAD or laryngoscopy attempts.

- Pediatric cricothyrotomy is a rare, age-appropriate, specifically authorized last-resort procedure for persistent cannot-oxygenate/cannot-ventilate failure.

Clinical pearls

- Do not wait for arbitrary attempt counts when the child is deteriorating - return to effective ventilation first.
- Waveform $EtCO_2$ confirms continuous respiratory gas exchange after an advanced airway and helps identify deterioration promptly.

Related pathways / contacts

AR-05 Pediatric Airway - Open for the primary pediatric airway and BVM pathway.

AR-08 Post-Intubation / BIAD Management - Open for airway confirmation, ventilation, monitoring, and continued post-airway management.

PC-02 Pediatric Bradycardia / Poor Perfusion - Open for HR below 60/min with poor perfusion despite oxygenation and ventilation.

PC-04 Pediatric Pulseless Arrest - Open if the child is pulseless.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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